Lecture 0:
Course Overview

"Clinical Engineering Course Overview & Key Concepts"

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Course Objectives

- Define the role of clinical engineers in healthcare.
- Manage medical technology across its life cycle.
- Ensure safety, regulations, and risk management.
- Integrate devices with IT and hospital infrastructure.
- Apply leadership, ethics, and professional standards.
- Adapt to emerging technologies in healthcare.

Why This Course Matters

- Healthcare today depends on technology at every level of care.
- Clinical engineers are the bridge between medicine and technology.
- Ensures the highest standards of patient safety, care quality, and operational efficiency within hospitals
- Prepares you for roles in device management, regulations, and innovation.
- Builds skills needed to face future challenges in health technology.
- High demand also exists in regulatory and quality management, which are strongly connected to clinical engineering.

Course Topics

1. Clinical Engineering Principles
2. Biomedical Instrumentation Fundamentals
3. Healthcare Technology Management
4. Safety and Regulatory Compliance
5. Clinical Informatics and IT Systems
6. Medical Facilities Engineering
7. Organizational and Project Management
8. Clinical Engineer Certification (CCE)

Teaching & Assessment Methods

1. *Lectures (Slides) , case studies*
2. *Assignments & small projects.*
3. *Midterm & final exam.*

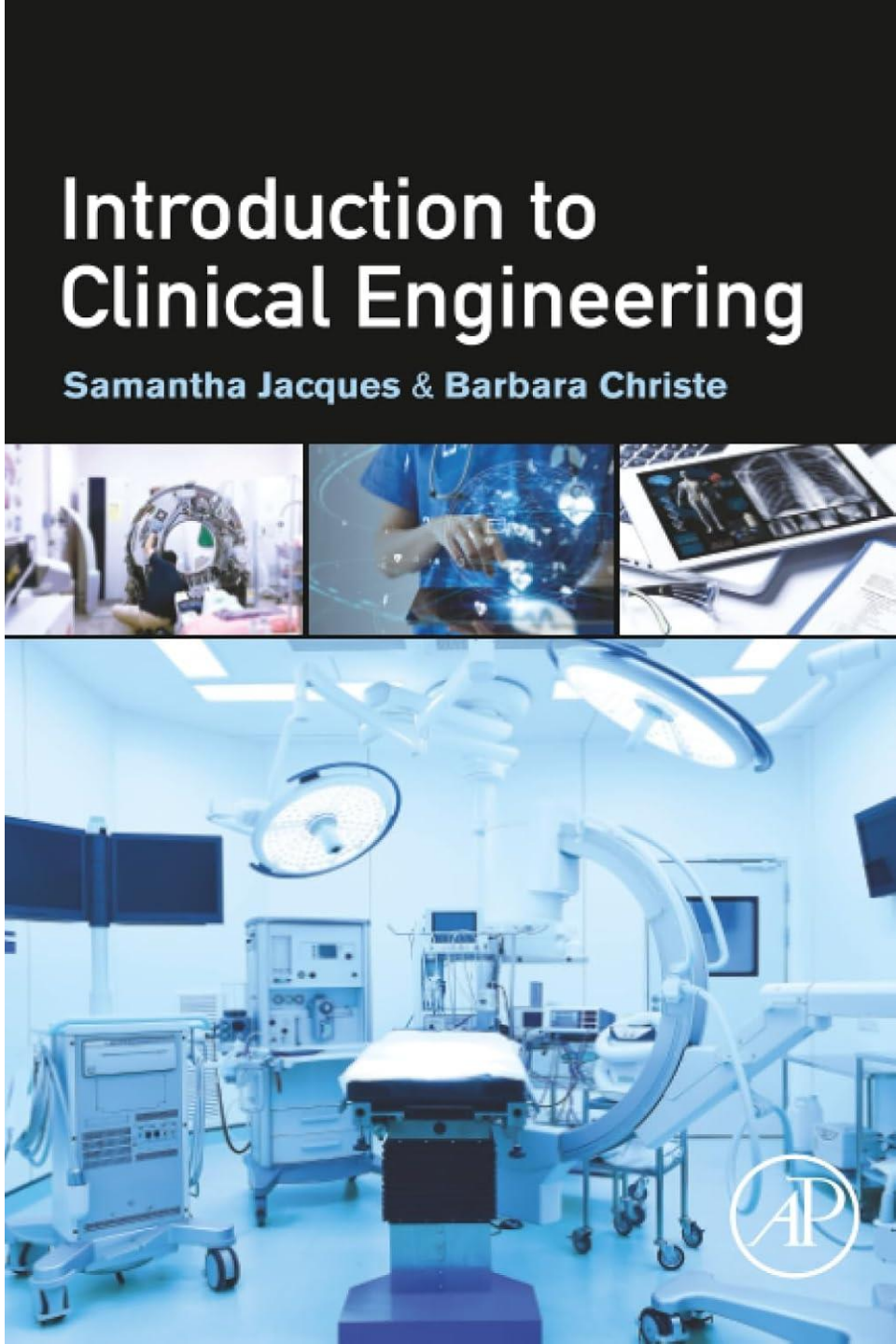
Teaching & Assessment Methods

Add

		Assessment Method	Time	Marks
حذف	تعديل	Homework Assignments		20%
حذف	تعديل	Mid-Term Exam	hour 1	20%
حذف	تعديل	Term Project		20%
حذف	تعديل	Final Exam	hours 2	40%

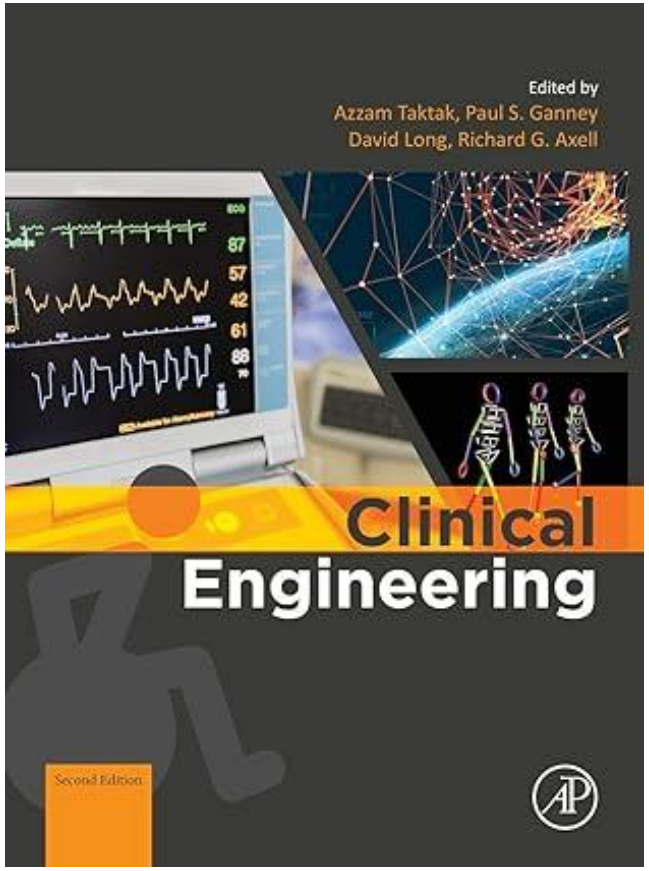
Main references

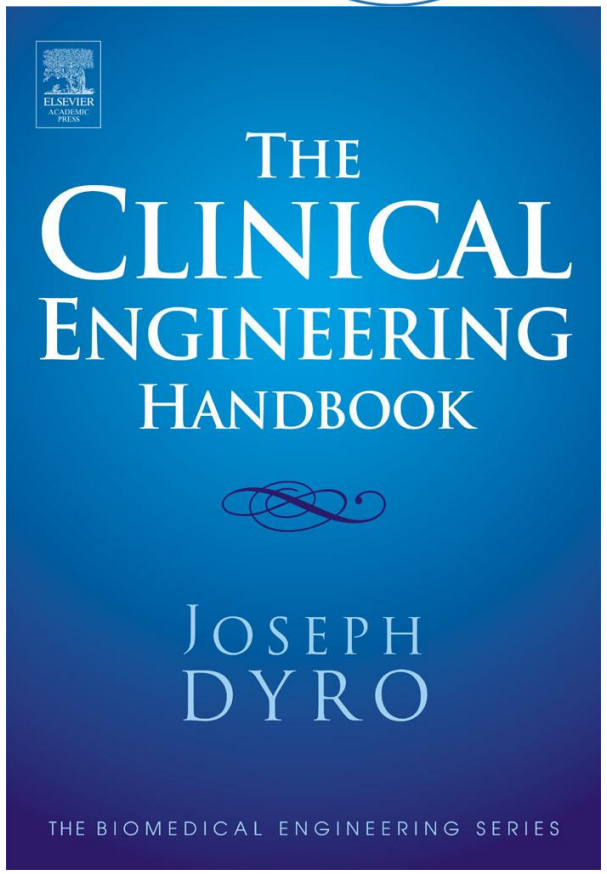
- SAMANTHA JACQUES & BARBARA CHRISTE. (2020). INTRODUCTION TO CLINICAL ENGINEERING. In *Introduction to Clinical Engineering* (pp. i–iii). Elsevier. https://doi.org/10.1016/B978-0-12-818103-4.00008-9



Othe Substantial References

- Taktak, A. F. G., Ganney, P., Long, D., & White, P. (Eds.). (2019). *Clinical engineering: A handbook for clinical and biomedical engineers* (Second edition). Amsterdam Waltham, MA: Elsevier : Academic Press.
- Dyro, J. (2004). *Clinical Engineering Handbook* (1st ed). San Diego: Elsevier Science & Technology.





Reference links

Main Related institutions:

- **ACCE – American College of Clinical Engineering (Use VPN)**

  https://accenet.org/CECertification/Pages/GetCertified.aspx
- **AAMI – Association for the Advancement of Medical Instrumentation**

  https://aami.org/
- **ASHE – American Society for Health Care Engineering**

  https://www.ashe.org/

Reference Certificates for the Clinical Engineering

- ACCE/HTCC → Certification as a clinical engineer (CCE) (engineers).

- AAMI/ACI → CBET & CHTM (technicians & managers).

Institutions & Certifications in Clinical Engineering

- **Main 3 (directly central):**
  - **ACCE** – American College of Clinical Engineering
  - **AAMI** – Association for the Advancement of Medical Instrumentation
  - **ASHE** – American Society for Health Care Engineering
- **Other 5 (supporting / related):**
  - **AHA** – American Hospital Association
  - **BMES** – Biomedical Engineering Society
  - **NFPA** – National Fire Protection Association
  - **NESCE** – New England Society of Clinical Engineering
  - **WHO** – World Health Organization
- **3 Key Certifications:**
  - **CCE** – Certified Clinical Engineer
  - **CBET** – Certified Biomedical Equipment Technician
  - **CHTM** – Certified Healthcare Technology Manager

Motivation & Closing

- Exciting emerging and prospects applications (AI in medical imaging, robotics in surgery, wearable devices , Regulatory paths).

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"Clinical Engineering is where technology meets patient care at health institutions — and you will play a part in shaping the future of healthcare."

Contact Information



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Think with me !? 🤔

- **Which topics do you expect to learn in this course?**
  - Beyond the objectives and topics we listed today…
  - What areas of clinical engineering interest you the most?
  - Is there a specific skill or knowledge you hope to gain?

- 💡 *Your input will help shape examples, discussions, and projects during the semester.*

Clinical Engineering

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Technical + Managerial + Patient Safety Focus


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Thank you